

Technology Stack & Technical Design

Cebuana Lhuillier — Parking Reservation System

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Audience: engineering, security, infrastructure

1. Executive Summary

The Parking Reservation System is a three-role corporate web application (End Users, Parking Attendants, Administrators) that manages reservations across multiple buildings, floors, and parking slots. It is built on a FastAPI backend and a React 19 single-page frontend, with a pluggable data layer that runs against either MongoDB 7.0 or Couchbase 7+ (Capella cloud or on-premise Enterprise) controlled by a single environment variable. The application is containerized for Docker / Kubernetes deployment and ships with a same-origin nginx reverse-proxy front end that eliminates browser CORS concerns and keeps HttpOnly auth cookies isolated to one origin.

Key architectural properties:

- Stateless backend (no in-process session affinity) — horizontally scalable
- All configuration via 12-factor environment variables; no hard-coded secrets
- Database abstraction layer with Motor-shaped API; switching DB engines requires zero call-site changes
- JWT (HS256) auth with HttpOnly Secure cookies and bcrypt password hashing
- Background coroutines for auto-no-show, waitlist expiry, and slot auto-release
- Rate limiting via SlowAPI (limits library) with per-IP buckets
- Persistent file uploads on a mounted volume (no GridFS dependency)
- Hardened CORS (fail-closed when CORS_ORIGINS env var is empty)
- VAPT-cleared and QAT-cleared on the most recent test cycles

2. High-Level Architecture

Three-tier architecture: a React 19 SPA in the browser; an asynchronous FastAPI 0.110.1 service tier mediating all business logic, authentication, and persistence; and a polyglot persistence tier where any deployment can choose between MongoDB 7.0 and Couchbase 7+ at boot.

Logical request flow

Browser → nginx 1.27.5 (TLS termination + same-origin `/api` proxy) → Uvicorn 0.25.0 (ASGI) → FastAPI 0.110.1 routers → Database Abstraction Layer → MongoDB 7.0.31 (via Motor 3.3.1) OR Couchbase 7.6.10-8025-enterprise (via couchbase 4.6.0).

Background workers (asyncio tasks)

Task	Cadence	Purpose
auto_mark_no_shows()	Every 60 s	Marks a reservation as no_show when its window has passed without check-in.
check_waitlist_expiry()	Every 60 s	Expires unresponded waitlist offers after the configured notification window.
auto_release_slots()	Every 60 s	Resets parking_slots.status to 'available' once the building's release_time has passed and no active reservation occupies the slot.

3. Backend Stack (Python)

Runtime & build

Component	Exact Version	Notes
Python	3.11.15	CPython, Linux x86_64
Uvicorn	0.25.0	ASGI server. Started with <code>--proxy-headers --forwarded-allow-ips='*'</code>
Docker base image (backend)	python:3.11.15-slim	Multi-stage Dockerfile; runs as non-root uid 10001

Core web framework & validation

Component	Exact Version	Purpose
FastAPI	0.110.1	ASGI web framework
Starlette	0.37.2	Underlying ASGI toolkit
Pydantic	2.12.5	Request/response models
pydantic_core	2.41.5	Rust-backed engine
email-validator	2.3.0	Backs EmailStr
python-multipart	0.0.22	Multipart form parsing
python-dotenv	1.2.1	.env loader for dev
aiofiles	25.1.0	Async file IO

Authentication & security

Component	Exact Version	Purpose
PyJWT	2.11.0	JWT HS256
passlib	1.7.4	Password hash framework
bcrypt	4.1.3	Hash primitive
python-jose	3.5.0	JWS/JWE alt
cryptography	46.0.4	TLS / signing primitives
slowapi	0.1.9	Rate limiting middleware
limits	5.8.0	SlowAPI storage

Data layer

Component	Exact Version	Purpose
Motor	3.3.1	Async MongoDB driver
PyMongo	4.5.0	Required by Motor
dnspython	2.8.0	SRV record resolution
couchbase	4.6.0	Couchbase Python SDK

AI / LLM & integration

Component	Exact Version	Purpose
emergentintegrations	0.1.0	Universal LLM key client
openai	1.99.9	OpenAI SDK
litellm	1.80.0	Multi-provider router
google-genai	1.62.0	Gemini SDK
google-generativeai	0.8.6	Gemini SDK (legacy)
tiktoken	0.12.0	Token counting
stripe	14.3.0	Payments SDK
boto3	1.42.42	AWS SDK
requests	2.32.5	Sync HTTP client
httpx	0.28.1	Async HTTP client

Document generation & utilities

Component	Exact Version	Purpose
openpyxl	3.1.5	Excel generation
python-docx	1.2.0	Word generation
weasyprint	68.1	PDF generation (this document)
Jinja2	3.1.6	Template engine
Pillow	12.1.0	Image handling
qrcode	8.2	QR codes for check-in
pandas	3.0.0	CSV bulk-upload + report exports
numpy	2.4.2	Pandas dependency
rich	14.3.2	CLI logging (dev)

Testing & developer tooling

Component	Exact Version	Purpose
pytest	9.0.2	Unit + integration
playwright	1.58.0	End-to-end browser automation
black	26.1.0	Formatter
flake8	7.3.0	Linter
isort	7.0.0	Import sorter
mypy	1.19.1	Static type checker

4. Frontend Stack (Node / React)

Runtime & build

Component	Exact Version	Notes
Node.js	20.20.2	Build-time only
Yarn	1.22.22	Package manager (yarn.lock = source of truth)
Docker base (build stage)	node:20.20.2-alpine	Stage 1 of Dockerfile
Docker base (runtime stage)	nginx:1.27.5-alpine	Stage 2 — serves static + reverse-proxies /api

Core framework

Component	Exact Version	Purpose
react	19.2.3	UI library
react-dom	19.2.3	DOM renderer
react-router-dom	7.13.1	Client-side routing (CVE-2025-43865 cleared)
react-scripts	5.0.1	CRA build pipeline
@craco/craco	7.1.0	Config override

UI primitives — Radix UI (shadcn/ui foundation)

Component	Exact Version
@radix-ui/react-accordion	1.2.12
@radix-ui/react-alert-dialog	1.1.15
@radix-ui/react-aspect-ratio	1.1.8
@radix-ui/react-avatar	1.1.11
@radix-ui/react-checkbox	1.3.3
@radix-ui/react-collapsible	1.1.12
@radix-ui/react-context-menu	2.2.16
@radix-ui/react-dialog	1.1.15
@radix-ui/react-dropdown-menu	2.1.16
@radix-ui/react-hover-card	1.1.15
@radix-ui/react-label	2.1.8
@radix-ui/react-menubar	1.1.16
@radix-ui/react-navigation-menu	1.2.14
@radix-ui/react-popover	1.1.15
@radix-ui/react-progress	1.1.8
@radix-ui/react-radio-group	1.3.8
@radix-ui/react-scroll-area	1.2.10
@radix-ui/react-select	2.2.6
@radix-ui/react-separator	1.1.8
@radix-ui/react-slider	1.3.6
@radix-ui/react-slot	1.2.4
@radix-ui/react-switch	1.2.6
@radix-ui/react-tabs	1.1.13
@radix-ui/react-toast	1.2.15
@radix-ui/react-toggle	1.1.10
@radix-ui/react-toggle-group	1.1.11
@radix-ui/react-tooltip	1.2.8

All shadcn/ui components (Button, Card, Dialog, Tooltip, Sonner, etc.) sit on the above Radix primitives.

Styling, forms, data viz, utilities

Component	Exact Version	Purpose
tailwindcss	3.4.19	Utility-first CSS
tailwind-merge	3.4.0	Class de-duplication
tailwindcss-animate	1.0.7	Animation utilities
class-variance-authority	0.7.1	shadcn variant system
clsx	2.1.1	Conditional className
postcss	8.5.6	CSS processor
autoprefixer	10.4.23	Vendor prefixing
react-hook-form	7.69.0	Form state
@hookform/resolvers	5.2.2	RHF + zod bridge
zod	3.25.76	Schema validation
recharts	3.6.0	Charts
axios	1.13.2	HTTP client
sonner	2.0.7	Toasts
date-fns	4.1.0	Date utilities
react-day-picker	8.10.1	Date range picker
react-markdown	10.1.0	Markdown renderer
react-resizable-panels	3.0.6	Panels
lucide-react	0.507.0	Icons
html5-qrcode	2.3.8	QR scan
input-otp	1.4.2	OTP input
embla-carousel-react	8.6.0	Carousel
vaul	1.1.2	Drawer/sheet
cmdk	1.1.1	Command palette
next-themes	0.4.6	Theme switcher

Linting & dev tooling

Component	Exact Version
eslint	9.23.0
@eslint/js	9.23.0
eslint-plugin-import	2.31.0
eslint-plugin-jsx-a11y	6.10.2
eslint-plugin-react	7.37.4
eslint-plugin-react-hooks	5.2.0
globals	15.15.0

5. Database Layer

Two interchangeable storage engines sit behind a shape-compatible Motor-style abstraction. Switching is controlled by the `DB_TYPE` environment variable; no application code needs to change.

Engine	Exact Version	Used When	Notes
MongoDB Server (CE)	7.0.31	DB_TYPE=mongodb (default)	Single-node Docker deployment uses official mongo:7.0.31 image; clusters use a managed service or replica set
Couchbase Server (Enterprise) on Capella	7.6.10-8025-enterprise	DB_TYPE=couchbase	18 native scopes-and-collections under the configured bucket; one Couchbase collection per logical entity

Logical collections (identical names on both engines)

users, sessions, buildings, floors, parking_slots, vehicles, zones, parking_configs, building_policies, slot_registrations, reservations, waitlist_entries, event_blocks, notifications, site_content, templates, ai_insights, migrations.

Indexes (Mongo and Couchbase parity)

- users.email (unique), users.id (unique), users.role
- reservations.id (unique), reservations.user_id, reservations.building_id, reservations.qr_token; compound (slot_id, date, status)
- buildings.id, floors.building_id, parking_slots.floor_id
- notifications compound (user_id, created_at DESC)
- waitlist_entries compound (building_id, preferred_date, status); compound (user_id, status)
- slot_registrations compound (slot_id, status); compound (user_id, status); compound (building_id, status)

- event_blocks compound (building_id, date)
- site_content.key (unique), parking_configs.building_id (unique), building_policies.building_id (unique)

6. Network, Web Server, & Infrastructure

Component	Exact Version	Role
nginx	1.27.5 (alpine)	Frontend container web server. Serves static React bundle, gzips text assets, reverse-proxies /api/*, sets X-Content-Type-Options/X-Frame-Options/Referrer-Policy
Docker Compose CLI	v2 (no version key required in compose file)	/app/docker-compose.yml; profiles {default, couchbase}
Kubernetes ingress	Per-deployment (NGINX ingress, Traefik, or LB)	Out-of-scope for this stack doc; TLS termination + edge caching live there

Container topology (docker-compose default profile)

Service	Image	Internal Port	Notes
frontend	parking-frontend:latest (built from /app/frontend/Dockerfile)	80	Multi-stage: yarn build → nginx; healthcheck on /healthz
backend	parking-backend:latest (built from /app/backend/Dockerfile)	8001	Uvicorn ASGI; healthcheck on /health; non-root user
mongo	mongo:7.0.31	27017	Optional — only when DB_TYPE=mongodb; mongo_data volume

7. Security Posture

Control	Implementation
Transport security	TLS terminated upstream (Caddy / Traefik / cloud LB). Backend honours X-Forwarded-Proto via uvicorn --proxy-headers.
Auth tokens	JWT HS256, signed with a 64-byte JWT_SECRET, delivered via HttpOnly Secure cookies (Secure flag toggled by COOKIE_SECURE env var).
Password storage	bcrypt 4.1.3 via passlib 1.7.4.
Brute force	SlowAPI rate limiter on /api/auth/login (per IP).
CORS	Fail-closed: <code>allow_origins</code> defaults to []. Operators must opt in via CORS_ORIGINS env var.
CSV/Excel formula injection	All free-text user fields sanitized at write time (single-user POST/PUT and bulk upload). Backfill script available.
Source maps	GENERATE_SOURCEMAP=false in frontend build.
API version disclosure	/api/ root returns only {message, status}; no version string.
Build-info verification	GET /api/system/build-info returns deploy timestamp + key package versions, unauthenticated, for cache-freshness retests.
CSV bulk upload privilege guard	Bulk upload silently downgrades any 'admin' role from CSV to 'user'.
Field-level access control	Non-admin reads of /api/parking-config and /api/slot-registrations are trimmed/scoped (V-02 fix).
Logout & session invalidation	Sessions persisted server-side; revoked on logout and on password change.

8. Build & Deploy Process

Backend image build

- Base: `python:3.11.15-slim`
- apt: build-essential, libssl-dev, ca-certificates, curl
- `pip install -r requirements.txt`
- `pip install emergentintegrations==0.1.0 --extra-index-url https://d33sy5i8bnduwe.cloudfront.net/simple/`
- Add non-root user uid=10001
- EXPOSE 8001 / CMD `uvicorn server:app --host 0.0.0.0 --port 8001 --proxy-headers --forwarded-allow-ips=*`
- HEALTHCHECK GET /health every 30 s

Frontend image build

- Stage 1 (build): `node:20.20.2-alpine` , `yarn install --frozen-lockfile` , `yarn build`
(`REACT_APP_BACKEND_URL` passed in as a build arg, may be empty for same-origin)
- Stage 2 (runtime): `nginx:1.27.5-alpine` , copies `/app/build` into `/usr/share/nginx/html` , custom `/etc/nginx/conf.d/app.conf` with the SPA fallback + /api reverse proxy
- EXPOSE 80
- HEALTHCHECK `wget /healthz` every 30 s

9. Environment Variables — Canonical List

Variable	Required?	Used By	Description / Default
DB_TYPE	Yes	Backend	'mongodb' (default) or 'couchbase' — selects active adapter
MONGO_URL	Conditional	Backend	Required when DB_TYPE=mongodb.
DB_NAME	Conditional	Backend	MongoDB database name. Default 'parking'
COUCHBASE_CONNECTION_STRING	Conditional	Backend	couchbases://... required when DB_TYPE=couchbase
COUCHBASE_BUCKET	Conditional	Backend	Capella/Enterprise bucket name
COUCHBASE_USERNAME	Conditional	Backend	DB user with bucket+collection mgmt permission
COUCHBASE_PASSWORD	Conditional	Backend	Paired credential
JWT_SECRET	Yes	Backend	64-byte hex string. Generate with: <code>openssl rand -hex 64</code>
JWT_ALGORITHM	No	Backend	Default HS256
JWT_EXPIRATION_HOURS	No	Backend	Default 0.5 (30 minutes)
COOKIE_SECURE	Yes (prod)	Backend	'true' = Secure flag set on auth cookie; 'false' for local plain HTTP
CORS_ORIGINS	No	Backend	Comma-separated list of allowed origins. Empty = fail-closed
UPLOAD_DIR	No	Backend	Path for uploaded layout images. Default /var/lib/parking/uploads
EMERGENT_LLM_KEY	Optional	Backend	Universal LLM key for AI Insights
REACT_APP_BACKEND_URL	Build-time	Frontend	Baked into JS bundle. Empty = same-origin proxy
FRONTEND_PORT	No	docker-compose	Host-side port mapping. Default 3000

10. Versioning & Upgrade Cadence

All Python dependencies are pinned to exact versions in `/app/backend/requirements.txt`. All Node dependencies use exact resolved versions captured in `/app/frontend/yarn.lock`. Reproducible builds are guaranteed by these two files.

Component	Cadence	Trigger / Notes
Security patches (any pinned dep)	Within 7 days of CVE disclosure	Subscribe to Pydantic, FastAPI, react-router-dom, bcrypt, cryptography advisories
Minor updates (Pydantic, FastAPI, React Radix)	Quarterly	Run pytest + Playwright suites against staging before promoting
Major upgrades (Python, Node, React, MongoDB, Couchbase)	Annually	Plan a 1-week stabilization window; verify both DB adapters
Container base images	Monthly rebuild	Picks up patched apt + alpine packages even when application code is unchanged

11. Repository Reference Map

File / Folder	Purpose
/app/backend/requirements.txt	Pinned Python dependencies
/app/backend/Dockerfile	Backend container build
/app/backend/server.py	FastAPI app, route mounting, CORS, startup tasks
/app/backend/database/	Abstraction layer (interface, mongodb, couchbase_db, __init__)
/app/backend/routes/	All API routers (auth, users, buildings, reservations, ...)
/app/backend/services/background.py	auto_mark_no_shows, check_waitlist_expiry, auto_release_slots
/app/backend/scripts/ provision_couchbase_collections.py	Idempotent Couchbase scope+collection provisioning
/app/backend/scripts/migrate_mongo_to_couchbase.py	Mongo → Couchbase data migration
/app/backend/scripts/backfill_csv_sanitization.py	V-08 backfill
/app/frontend/package.json + yarn.lock	Frontend dependency manifest + lockfile
/app/frontend/Dockerfile	Multi-stage frontend build
/app/frontend/nginx.conf	Same-origin reverse proxy + SPA fallback
/app/docker-compose.yml	Three-service stack: backend + frontend + mongo (profile-gated)
/app/.env.example	Canonical env-var template
/app/docs/DEPLOYMENT.md	Step-by-step deployment guide
/app/docs/output/VAPT_Tracker_2026-03-25.xlsx	Latest VAPT remediation tracker
/app/docs/output/ QAT_Failed_Items_Tracker_2026-03-23.xlsx	Latest QAT failed-items tracker
/app/memory/PRD.md	Living product requirements doc